

Keyu (Chloe) Cai

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WORK EXPERIENCE

AARP

Washington, DC

Data Science Intern

June 2024 – December 2024

- **Explorative Data Analysis & Customer Segmentation:** Supported digital transformation initiatives by collecting and analyzing communication history, user behavior, and conversion date, conducting customer segmentation across 23M+ members with **K-Means** and **t-SNE** using **PySpark**, improving the customer digital migration efficiency.
- **Data Pipeline:** Built an automated customer segmentation pipeline in **Databricks** to streamline feature extraction, data preprocessing, clustering, and **LLM-generated reporting**, reducing manual workload and enhancing decision-making for the analytics and marketing teams.
- **Model Development:** Trained and implemented a hybrid modeling approach by combining **Linear Regression** and **XGBoost**, predicting customer propensity towards digital transformation, improving targeting accuracy by 15% and increasing conversion rates by 2%.
- **A/B Testing:** Performed pseudo-**A/B test** on a 70K-participant email-opted-in campaign, evaluating renewal rates across different incentive structures and communication strategies, optimizing retention strategies.
- **Online Model Monitoring:** Developed an automated **lift curve** monitoring template in **Databricks** to track email-click-model performance, detecting performance degradation and triggering model recalibration, ensuring sustained predictive accuracy.

SELECTED PROJECTS

A House Subleasing Database System Design with Flask

January 2024 – May 2024

Optimized data management and system efficiency by designing and developing a web-based database system for house subleasing.

- Led quality database architecture by creating conceptual and logical designs using **UML/ER diagrams** and **relational schemas**.
- Developed the website application utilizing **HTML** and **JavaScript** for the frontend, **Flask** as the web framework, and **MySQL** as the database engine.
- Directed seamless deployment and scalability, conducting system test on a virtual machine hosted by **Google Cloud Platform**.
- Optimized response times and improved efficiency by 5% through implementation of **clustered indexing**.

Text Classification Using Machine Learning Methods and LLaMA

February 2024 – May 2024

Enhanced information retrieval and event monitoring by improving machine learning models for automated text classification.

- Preprocessed and structured a dataset of 20,000+ news articles across 20 categories by applying text tokenization, stop-word removal, and **TF-IDF** weighting, improving data quality for classification models.
- Trained and evaluated baseline classification models (**Naïve Bayes**, **Logistic Regression**, **Text CNN**, **RNN**, **BERT**), identifying BERT as the best performing model and fine-tuning it to achieve 95.4% test accuracy.
- Applied **bootstrapped resampling** to balance class distributions, enhancing model robustness and reducing bias.
- Leveraged **LLaMA** as Teacher Model for knowledge distillation, generating soft labels to optimize BERT predictions, improving classification accuracy by 2.5% while reducing computational cost, resulting in a lightweight BERT model for scalable text classification tasks.

EDUCATION

M.S. in Information Management | GPA: 3.96/4.00

May 2025

University of Illinois, Urbana-Champaign

- Machine Learning, Database Systems, Data Mining, Information Modeling, Data Warehousing and BI

M.S. in Environmental Engineering | GPA: 3.97/4.00

December 2022

Graduate Certificate in Applied Data Science

University of California, Berkeley

- Applications in Data Analysis, Computational Structures in Data Science, Principles and Techniques of Data Science

B.E. in Environmental Engineering | GPA: 4.34/5.00

July 2019

Tongji University

SKILLS

Programming Language: Python, SQL, R, MATLAB, Java, JavaScript, HTML, CSS, GO, C++, C#

Tools and Frameworks: Pandas, NumPy, statsmodels, PyTorch, PySpark, Apache Spark, PowerBI, Tableau, Scikit-learn, nltk, TensorFlow, keras, TF-IDF, Transformers, XGBoost, BERT, LSTM, Matplotlib, Seaborn, Git, Flask, Neo4j, MongoDB

Certificates: AWS Certified Cloud Practitioner, MySQL 8.0 Database Developer